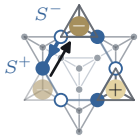


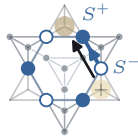
i.a) create


 $S_i^+ S_j^-$: pair splits

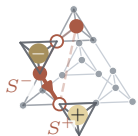
i.b) move


 $S_i^+ S_j^-$: spinon walks

i.c) annihilate


 $S_i^+ S_j^-$: + meets -
 $\mathbf{q}_\gamma = (0, 0, 0)$

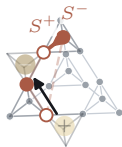
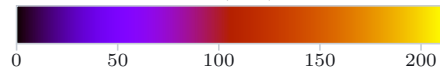
i.d) create


 $S_i^+ S_j^-$: pair splits + hops across the boundary they meet: $\mathbf{q}_\gamma = (0, 0, -1)$

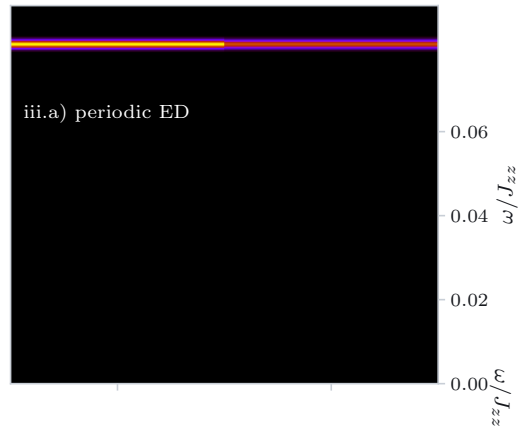
i.e) transport



i.f) annihilate


 $S^{zz}(\mathbf{q}, \omega)$


iii.a) periodic ED



iii.b) winding-free

